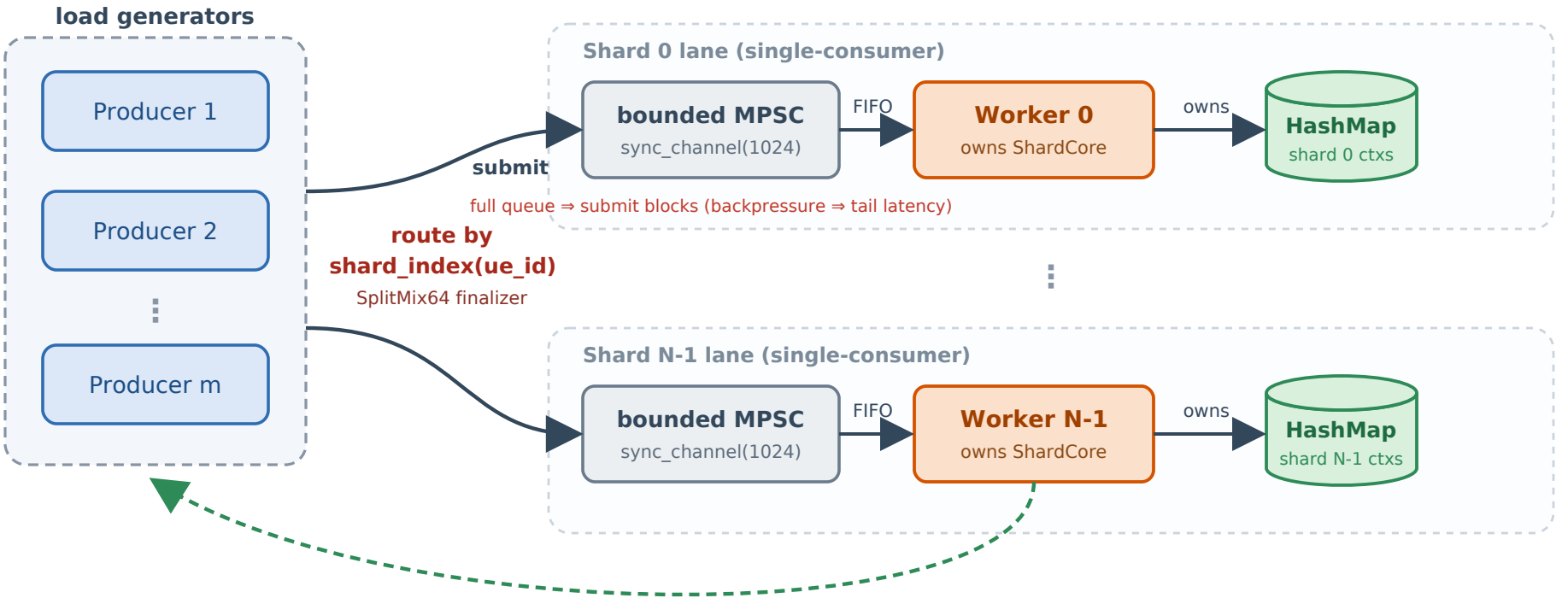
Shared-nothing, thread-per-shard architecture

Each shard is owned by exactly one worker — no shared mutable state, no locks on context data (spec §7)